Supplementary Simulation Tables (SOCIOMAP)

Supplementary Table S1: Global simulation configuration (default DGP + pipeline settings)

Default values in SimConfig (unless overridden by a scenario in Table S3).

Category	Parameter (code)	Default	Meaning
Cohort size	N	44,927	Individuals per replicate
Social structure	K_tribe	100	Number of ethnolinguistic groups ("tribes")
	K_region	3	Regions (used if include_region=True)
	K_religion	5	Religions
	include_region	True	Whether region is simulated and used
Tribe size distribution	enable_many_small_tribes	True	If True, tribe sizes are heavy-tailed
	zipf_a	1.2	Zipf exponent controlling many-small-groups behavior
Random effects	tribe_re_sd	0.25	SD of tribe effects in latent socioeconomic S and latent risk H
	region_re_sd	0.35	SD of region effects in latent socioeconomic S and latent risk H
Diagnosis access model	enable_access_bias	True	If True, underdiagnosis depends on SES/urbanicity
	access_intercept	-0.2	Baseline access logit
	access_income_w	0.8	Weight for income in access model
	access_edu_w	0.7	Weight for education/3 in access model
	access_urban_w	0.6	Weight for urbanicity in access model

	access_region_sd	0.4	SD of region-specific access offsets
Missingness model	miss_bmi_base	0.05	Base probability BMI missing
	miss_bmi_lowSES_boost	0.15	Additive boost if income=0 (low SES)
	miss_bmi_rural_boost	0.08	Additive boost if urban=0 (rural)
	miss_income_base	0.03	Base probability income missing
	miss_income_region_boost	0.10	Additive boost if region==0
	miss_diet_base	0.04	Base probability diet variables missing
	miss_diet_lowEdu_boost	0.10	Additive boost if education <= 1
DGP toggles	enable_main_effect_heterogeneity	True	Adds main-effect heterogeneity via tribe/region shifts
	enable_interactions	True	Enables effect modification terms in disease logits
	enable_rare_disease	False	Adds "RareDiseaseX" if True
	enable_cluster_truth	True	Creates a ground-truth cluster label for ARI/NMI
	distribution_shift	False	If True, generates a second test cohort with shifted parameters
Inference safeguards	min_group_n	50	Min group size for chi-squared tests
	fdr_alpha	0.05	BH-FDR threshold for chi-squared p-values
Explainability	shap_top_k	15	Top-k features used in SHAP recovery metric
Unsupervised	umap_n_neighbors	30	UMAP neighborhood size
	umap_min_dist	0.1	UMAP min distance
	kmeans_k	5	Number of KMeans clusters

Supplementary Table S2: Disease "truth" specifications and ICD noise parameters

Exact values in default_diseases() for the base set.

Disease	alpha	lam	BMI hinge	Hinge coef	oil_income_gamma	bmi_tribe_gamma	ICD sens	ICD spec
Hypertension	-5.0	0.20	27.0	0.10	0.55	0.00	0.90	0.995
Type2Diabetes	-6.0	0.22	26.0	0.12	0.20	0.10	0.88	0.996
ChronicHepatitis	-5.7	0.10	28.0	0.04	0.10	0.00	0.80	0.993
Asthma	-6.3	0.08	30.0	0.02	0.00	0.00	0.85	0.996
Epilepsy	-7.2	0.05	29.0	0.01	0.00	0.00	0.78	0.997

Supplementary Table S2b: Covariate coefficients (coef dict) used by each disease

Each disease logit adds sum_k beta_k X_k with the values below.

Disease	income	education	urban	palm_oil	fruitveg	smoke	alcohol	phys_act
Hypertension	-0.10	-0.05	0.05	0.20	-0.08	0.08	0.04	-0.08
Type2Diabetes	0.05	-0.02	0.08	0.10	-0.06	0.04	0.02	-0.10
ChronicHepatitis	-0.12	-0.04	-0.03	0.12	-0.03	0.05	0.10	-0.02
Asthma	-0.03	-0.02	0.10	0.02	-0.01	0.12	0.01	-0.01
Epilepsy	-0.08	-0.06	-0.05	0.01	-0.01	0.02	0.02	-0.01

Supplementary Table S3: Scenario grid (overrides relative to defaults)

Documents what each scenario changes in make_scenarios().

Scenario	Purpose	Key overrides (relative to defaults)

S1_null	Sanity check: no structure	enable_main_effect_heterogeneity=False; enable_interactions=False; enable_access_bias=False; enable_cluster_truth=False
S2_main_effects	Main-effect heterogeneity only	enable_interactions=False; enable_access_bias=False; enable_cluster_truth=True
S3_interactions	Interactions present (effect modification)	enable_interactions=True; enable_access_bias=False; enable_cluster_truth=True
S4_access_bias	Access-driven underdiagnosis dominates	enable_main_effect_heterogeneity=False; enable_interactions=False; enable_access_bias=True; enable_cluster_truth=False; access_intercept=-1.2; access_income_w=1.0; access_edu_w=1.0; access_urban_w=0.8
S5_many_small_tribes	Small-cell stress test	K_tribe=150; zipf_a=1.6; min_group_n=80; enable_interactions=True; enable_access_bias=True; enable_cluster_truth=True
S6_rare_disease	Severe class imbalance	enable_rare_disease=True (adds RareDiseaseX); enable_interactions=True; enable_access_bias=True; enable_cluster_truth=True
S7_multimorbidity_strong	Stronger shared-factor structure	tribe_re_sd=0.35; region_re_sd=0.45; enable_interactions=True; enable_access_bias=True; enable_cluster_truth=True
S8_distribution_shift	Generalization under shift	distribution_shift=True (creates a second cohort with: zipf_a=max(0.9, zipf_a-0.3); access_intercept=access_intercept-0.2)

Supplementary Table S4: RareDiseaseX specification (only in S6)

Exact parameters in default_diseases(enable_rare=True).

Disease	alpha	lam	BMI hinge	Hinge coef	oil_income_gamma	bmi_tribe_gamma	ICD sens	ICD spec
RareDisease	-9.0	0.1	27.0	0.08	0.10	0.05	0.75	0.99

X		0						8